

Safety, tolerability and clinical effects of rovonaptabin, also known as BC007 on fatigue and quality of life in patients with Post-COVID syndrome (reCOVER): a prospective, exploratory, placebo-controlled, double-blind, randomised phase IIa clinical trial (RCT)



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Summary

Background Rovonaptabin neutralises functional autoantibodies targeting G-Protein coupled receptors (GPCR-fAABs), observed in patients with Post-COVID syndrome. As we hypothesise an improvement of PCS by rovonaptabin, the aim of reCOVER was to investigate safety, tolerability, and clinical effects of rovonaptabin in PCS patients.

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Methods reCOVER is a prospective, exploratory, placebo-controlled, double-blind, randomised phase IIa clinical investigator initiated trial with 1350 mg rovonaptabin with additional cross-over at the Universitätsklinikum Erlangen, Germany. The trial was registered in EudraCT, 2022-001781-35. Screening was done between 21.11.2023 and 25.06.2024. Eligible participants (18–80 years) showed GPCR-fAABs, at least 3/8 defined PCS symptoms persisting ≥ 3 months after COVID-19 and fatigue as major symptom. Participants were randomly assigned (1:1) to either receive rovonaptabin or placebo at day 0 (d0) and d48 with a follow-up of 28 days, respectively. Primary endpoint was the number of treatment emergent adverse events (TEAE) at d28 (co-primary endpoint: TEAE at d70); secondary endpoint focused on fatigue and quality of life.

Findings Thirty PCS patients were randomised and analysed. RCT analysis showed nine (rovonaptabin) and five TEAEs (placebo), yet without statistically significance ($p = 0.1299$; CI –14.80%; 63.02%); one serious adverse event, not related to treatment, was recorded. Rovonaptabin showed a neutralisation of GPCR-fAAB and a significant improvement of FACIT Fatigue Scale (effect size = 2.10, $p = 0.0378$), Bell score (effect size = 3.64, $p = 0.0004$), Fatigue Severity Scale (effect size = –2.66, $p = 0.0088$), and quality of life (4/8 items).

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Interpretation As this proof-of-concept study showed effects on the patient-centred endpoint PCS and a good safety profil, subsequent studies are needed to confirm these results in a larger cohort.

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Research in context

Evidence before this study

Post-COVID syndrome, also known as post-acute sequelae of SARS-CoV-2 (PASC) or Long-COVID (LC), is defined as the persistence of clinical symptoms longer than three months after acute COVID-19. Currently, PCS therapy is based on a multidisciplinary approach with symptomatic, yet not causal treatment. Recent data suggest that PCS encompasses different subgroups, each of them requiring a specific causal therapy. One of those subgroups is defined by an autoimmune phenotype with a seropositivity of functional autoantibodies targeting G-protein coupled receptors (GPCR-fAAb). The aptamer rovnaptabin (BC007) has been shown to neutralise such GPCR-fAAb *in-vitro* and *in-vivo*, therefore offering a causal therapeutic potential for PCS patients with GPCR-fAAb. We searched PubMed from inception until December 12th, 2024, for articles with the terms “BC007”, “BC 007”, “rovnaptabin”, “Post-COVID RCT”, and “PASC RCT” with no language restriction. This identified 53 articles, only one of which dealt with the use of rovnaptabin in PCS patients in the form of an individual healing attempt of a single PCS patient, performed by our group. Another clinical phase II trial investigating the treatment of PCS patients with rovnaptabin (BLOC, ID: NCT05911009) has been registered

in the database clinicaltrials.gov, however no publication has been issued yet.

Added value of this study

The reCOVer study is the first prospective, exploratory, placebo-controlled, double-blind, randomised phase IIa clinical investigator-initiated trial (IIT) of rovnaptabin (BC007) with additional cross-over in patients with Post-COVID syndrome. This is the first investigator initiated clinical study to investigate rovnaptabin within this population. Rovunaptabin showed a temporary expected anticoagulatory effect with a good safety profile. Using complementary patient-reported outcome measurements, rovnaptabin showed a significant improvement of fatigue and its severity and quality of life in PCS patients compared to placebo.

Implications of all the available evidence

Due to the high disease burden and the yet unmet need of a causal therapy for PCS patients, the data of reCOVer indicate that rovnaptabin might offer a novel therapeutic option for an autoimmune subgroup of PCS patients. The data of the present study provides support for further clinical studies in a larger patient group with a long-term follow-up.

Introduction

Post-COVID syndrome, also known as post-acute sequelae of SARS-CoV-2 (PASC) or Long-COVID (LC), summarises clinical symptoms, persisting longer than three months after acute Coronavirus disease-19 (COVID-19).¹ Due to this complex condition with potentially multiple and overlapping causes patients' symptoms vary widely and only partially correlate with the severity of acute infection.² Prevalence differs depending on hospitalisation during acute COVID-19 and vaccination status.³ The exact pathogenesis of PCS is still elusive, yet different pathogenetic molecular pathways, indicating diverse PCS subgroups are proposed (e.g., long-term organ dysfunction, viral persistence, endothelial dysfunction, clotting phenomena).^{4–7} Of interest, an autoimmune subgroup of PCS patients shows different autoantibodies, targeting G-protein coupled receptors, with functional activity (GPCR-

fAAb).⁸ There are no systematic studies of the prevalence of GPCR-fAAb in large populations yet. Neutralisation of those GPCR-fAAb by rovnaptabin, also known as BC007 (Berlin Cures GmbH, Berlin, Germany), was accompanied by an improvement of capillary microcirculation and PCS patient's symptoms in an individual healing attempt performed by our group.⁹ With the hypothesis that neutralisation of GPCR-fAAb by rovnaptabin could increase capillary microcirculation and improve clinical PCS symptoms, we conducted a prospective, exploratory, placebo-controlled, double-blinded, randomised phase IIa clinical trial of rovnaptabin with additional cross-over afterwards in an autoimmune subgroup of PCS patients (reCOVer). Rovunaptabin is a 15-mer single strand DNA oligonucleotide, thrombin binding aptamer,¹⁰ later shown to bind and neutralise various GPCR-fAAb *in-vitro* and *in-vivo*.¹¹ Clinical data of a

phase I study (NCT02955420) showed a good safety profile in healthy participants with a mild-to-moderate increase in activated partial thromboplastin time (aPTT).¹² Here we report on safety, tolerability and clinical effects of rovonaptabin on fatigue and quality of life in PCS patients.

Methods

Trial design

The prospective, controlled, double-blind phase IIa randomised clinical trial with additional cross-over design afterwards of rovonaptabin (reCOVER) was conducted at the Department of Ophthalmology, Universitätsklinikum Erlangen (UKER), Erlangen, Germany as an investigator-initiated trial (IIT).

A total of 37 PCS patients were selected for screening according to the inclusion and exclusion criteria by the investigators. Adult female and male patients, between ages of 18–80 years were eligible for inclusion. Previous SARS-CoV-2 infection had to be documented by a positive polymerase chain reaction (PCR) test. PCS symptoms had to persist at least three months prior to study inclusion, with a clinically relevant fatigue (Bell score ≤ 60) and at least three of the following symptoms: exercise intolerance, exertional dyspnoea, concentration disorders, brain fog, fatigue, unsteady gait, disorders of the sense of taste and/or smell, headaches. Furthermore, the presence of GPCR-fAAs was a prerequisite for participation. Exclusion criteria were: the therapeutic use of anticoagulants that could interact with the investigational medicinal product (IMP) rovonaptabin and increase the tendency to bleed in the last two weeks before the screening visit; signs of long-term COVID-19 induced organ damage; a history of other known illnesses that could be cause for fatigue and reduced resilience (e.g., multiple sclerosis, anaemia); a history of obstructive sleep apnoea syndrome; obesity with a Body-Mass-Index ≥ 30 kg/m², acute and severe organ damage to the heart and lungs, malignancies or remission of malignant diseases within the last two years. No intake of interfering drugs that could lead to fatigue (e.g., opiates, antihistamines, antidepressants, NMDA antagonists) was permitted. Pregnant and breastfeeding persons were excluded. Furthermore, eye diseases with impairment of the retinal microcirculation, systemic diseases with eye involvement and glaucoma were excluded as reCOVER was accompanied by additive eye-based diagnostic measurements (e.g., OCT-angiography) for visualisation of capillary microcirculation, flow cytometry for measurements of the physics of blood cells, 7 T MRT scans.

A total of 30 patients were included in the study and randomised (1:1). The duration of participation was 91 days per participant according to the study protocol. The cross-over design was chosen to increase patients'

participation in the study (i.e., each participant will receive rovonaptabin). As no causal therapy is available in PCS treatment, no specific washout phase of other drugs was required prior to starting the trial (comorbidities can be seen in [Suppl. Tables S1 and S2](#), concurrent medication in [Suppl. Table S3](#)). For this initial exploratory pilot study, a sample size of $n = 30$ was considered sufficient, to obtain data for a subsequent study with an effect size of 0.35.¹³ The clinical trial was approved by the German Competent Authority (Federal Institute for Drugs and Medical Devices, BfArM) and the ethics committee of the Friedrich-Alexander University Erlangen-Nürnberg (23-50-Az). It is registered in the European Union Drug Regulating Authorities Clinical Trials Database (EudraCT No.: 2022-001781-35). Prior to enrolment the trial was registered in the German Clinical Trials Register (DRKS00032237, <https://drks.de/search/de/trial/DRKS00032237>, the trial results are not yet published at DRKS), providing the protocol and statistical analysis plan. The clinical trial was conducted in accordance with the study protocol, regulatory requirements, the ICH-GCP and the ethical principles of the Declaration of Helsinki. Written informed consent was obtained from all participants in the study. The clinical trial has been completed within 12 months on September 2024.

Randomisation and masking

The randomisation sequence was generated by the statistician before starting the enrolment with the maximal procedure with software R 4.3.0, a 1:1 ratio and a maximum tolerated imbalance of 6.¹⁴ The length of the randomisation sequence was 40 to account for possible screening failures. The study was performed in a double-blinded manner, where participants and investigators were blinded. Unblinded members of the clinical trial team packed the drug and performed statistical analysis. The blinding was not broken during the trial, as no medical emergency occurred. Unblinded persons of the pharmacy, being responsible for allocation, produced the ready-to-use infusion of rovonaptabin for intravenous application. Allocation concealment was provided by pharmacy of UKER according to the randomisation list. Consequently, 1350 mg rovonaptabin (in 50 ml NaCl 0.90%) or placebo were prepared and labelled in accordance with regulatory requirements. The carrier solution of 50 ml 0.90% sodium chloride, which had the same appearance as rovonaptabin, was used as placebo. In case of emergency unblinding envelopes were stored at the study site.

Procedures

Eligible participants entered the randomised, double-blind, RCT (part 1), receiving either 1350 mg rovonaptabin or placebo at d0 (V2); afterwards a cross-over (part 2) at d42 (V8) was performed. The allocation was

generated by the Centre for Clinical Studies CCS (AStr), participant enrolment was done by the investigator (AS, TK, BH, VZ, ASk). Assignment to treatment was done by the pharmacy; no other study personnel had access to the allocation list. During the cross-over (part 2) patients received either 1350 mg rovonaptabin followed by placebo, or placebo, followed by 1350 mg rovonaptabin. Total study duration per participant was 91 days, including 14 visits (Suppl. Table S4): After screening (V1), first study treatment with either IMP or placebo was administered (V2, d0) with follow-up visits during the following two days, respectively (V3, V4). Further study visits were after one week (V5), two weeks (V6), and four weeks (V7). After a 14-day break the switch to the second treatment with IMP or placebo in the cross-over procedure began on d42 (V8). The study visits were conducted on d43 (V9), d44 (V10), d49 (V11), d56 (V12), d70 (V13), and d91 (V14 [EOS]). Prior to each treatment, the presence of a negative SARS-CoV-2 antigen test, not older than 24 h, was required. Treatment was done by intravenous application of 1350 mg rovonaptabin (in 50 ml NaCl 0.90%) within 75–90 min or placebo (50 ml NaCl 0.90% alone) within 75–90 min. No concomitant care was applied by the study team.

Safety

For analysis of safety and tolerability adverse events (AE) and vital signs (e.g., blood pressure) were recorded during the whole observation period. Haematology, blood-chemistry and coagulation tests were performed at all study visits. A physical examination was done on both treatment days (d0/d42), d1, d28, d43, and d70. Occurrence of AEs, adverse reaction (ARs), serious adverse event (SAEs) and serious adverse reaction (SARs) were recorded during the entire observation period, classified according to the Common Terminology Criteria for Adverse Events CTCAE (5 categories: mild-moderate-severe-life-threatening-fatal and coded using MedDRA v. 27.0). Serious adverse events were reported according to regulatory requirements. All relevant data were recorded in the eCRF.

Outcomes

Primary endpoint of the RCT was the occurrence of self-reported in face to face treatment-emergent adverse events (TEAEs) in the period from V2 (after administration of rovonaptabin) to d28 compared between treatment and placebo group. Secondary endpoints were the treatment effect of rovonaptabin on self-reported fatigue and self-reported quality of life, assessed by different scores: the Functional Assessment of Chronic Illness Therapy (FACIT) Fatigue Scale and Chalder Fatigue Scale (for fatigue), the Bell scale and Fatigue Severity Scale (FSS) for severity of fatigue, the Canadian Criteria (CCC, for the presence of myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS)), SF-36 questionnaire (for quality of life), DePaul

Symptom Questionnaire - Post-Exertional Malaise short form 8 (DSQ-PEM) and 6 min walking test (6MWT, Borg scales). The reply of each questionnaire was done individually from each participant locally at each visit and collected by blinded persons of the study team. The FACIT-Fatigue Scale is a self-reported 13-item questionnaire, assessing fatigue and its impact on daily activities over the past seven days. It is rated on a four-point Likert scale (4 = “not tired at all” to 0 = “very tired”). Rating is scored from 0 to 52 (healthy). The 11-item Chalder Fatigue Scale (CFS) assesses physical fatigue (questions 1–7) and mental fatigue (questions 8–11), scored with four response options ranging from ‘less’ to ‘much more’. Rating is scored from 0 (healthy) to 33. The Bell Scale assesses fatigue severity, comprising a total of 10 responses with a rating from 0 to 100 (healthy). The 9-item FSS scored from 1 (“disagree”) to 7 (“completely agree”). The lower scores, the healthier the participant is. CCC comprises seven categories, including fatigue, PEM, sleep dysfunction, pain neurological and cognitive, autonomic, neuroendocrine and immune manifestations for a binary diagnosis of ME/CFS. The 5-item DSQ-PEM is scored from 0 (not present) to severe (including grading from 1 (mild) to 4 (very severe)); additional five items were recorded in a binary and time-coded score. Quality of life was assessed by the 36-item Short Form (SF-36), including eight subscales, the physical component summary (PCS) and the mental component summary (MCS). Rating is scored from 0 to 100 (healthy). 6MWT registers walking distance in a self-determined speed within 6 min; data for persons, which were unable to perform the 6MWT, were excluded from this analysis. BORG rating of perceived exertion scale assessed the subjectively perceived exertion during 6MWT, graded from 6 (no exertion) to 20 (maximum exertion). BORG-dyspnoea-scale assessed the subjectively perceived dyspnoea with a scoring from 0 (no) to 10 (maximum). Fatigue scores and quality of life were assessed at all visits from d0 to d91 (except at V1, V9). Six minute walk test (6MWT) was performed at d0, d7, d28, d42, d49, d70, and d91. Additional analysis of the within-participant differences (part 2: cross-over) was conducted after RCT for the co-primary endpoint, i.e., the occurrence of TEAEs from baseline (after administration of rovonaptabin) to d70. Data on race and ethnicity were not collected this was not the focus of the present study.

Cardiomyocyte bioassay

GPCR-fAAb testing was done at screening, d7, d28, d49, and d70, respectively. For the detection of GPCR-fAABs, a cardiomyocyte (CM) bioassay was performed by G. Wallukat (Berlin Cures GmbH) as described previously.¹⁵ IgGs from patient sera were heat inactivated (56 °C), dialysed (dialysis buffer, a 0.15 M NaCl, 10 mM phosphate buffer, pH 7.40; Membra-Cel MD 44,

14 kDa, Serva, Heidelberg, Germany) and stored at -20°C . CMs were isolated from new-born Wistar rats and maintained in culture for a maximum of eight days. The IgG fraction was added to the cells in final dilution of 1:50 for 60 min following the determination of basal cell beating. Depending on the direction of the Δ frequency: positive chronotropic GPCR-fAABs (β 2-fAAB, AT1-fAAB, α 1-fAAB, Noci-fAAB) were identified with the following inhibitors: ICI118, 551 (0.10 μM), losartan (1 μM), urapidil (1 μM), and J113397-7 (0.10 μM), respectively; to determine negative chronotropic GPCR-fAABs (MAS-AAB, M2-AAB, ETA-AAB) the following inhibitors were used: A779 (1 μM), atropine (1 μM), and BQ123 (0.10 μM). All inhibitors were obtained from Sigma-Aldrich Chemie GmbH (Taufkirchen, Germany).

Statistical analysis

The statistical analysis was done by the software R.¹⁶ The statistical analysis was pre-specified in a statistical analysis plan, which was finalised before the start of the enrolment. The statistical analysis was based on a frequentist approach; significance level was set to 5% and tests and confidence intervals were two-sided. The study design was exploratory. For each analysis the population were those patients, who provided data for this analysis. Missing values were not imputed. Thus, each fatigue score was tested separately without correction for multiple testing, as we did not know which of the endpoint would show a potential effect after treatment. The primary variable was the relative frequency of patients with at least one TEAE in the period from baseline to d28. This frequency is calculated for both treatment groups and a 95%-confidence interval for the difference of these two frequencies is estimated.¹⁷ Also, the rate of TEAE per day was calculated and compared between the two treatment groups by a Poisson-test. The same methods were applied to the co-primary variable (TEAE from baseline to d70). Considering the different baseline values, baseline variable was a covariate. So, the models were adjusted for these imbalances. For all other variables the data analysis was split in two parts: data acquired RCT at visit 8 and for all data up to visit 13. RCT analysis was done by linear mixed models adjusted for the baseline variable. The linear mixed models with random effects for repeated measurements were used to compare difference between groups at d0, d28, and d70. We estimated the treatment effects between rovonaptabin and placebo for RCT data up to d28; for the cross-over data we used cross-over differences. Statistical methods for cross-over studies are described in Senn et al.¹⁸ Cross-over differences were estimated with linear models with random effects for repeated measurements and adjusted for baseline variables. For further details see [Supplementary Material](#).

Role of funding source

The funders of the study (German Federal Ministry of Education and Research, BMBF, 01EP2108A, reCOVER; 01EO2105, iIMMUNE_ACS; German Research Foundation, DFG; 401821119/GRK2504) had no role in study design and conduct, data collection, analysis and interpretation, and writing of the manuscript.

Results

Screening of the first patient was done on 22nd November 2023, screening of the last patient was screened on 25th June 2024. The final study visit was on 10th September 2024; the study protocol remained unchanged during the course of the study. 37 patients with PCS were screened and assessed for eligibility, of whom 30 could be recruited for participation ([Fig. 1](#)). Reasons for exclusion were declined participation and withdrawal of consent ($n = 4$), not fulfilling either the inclusion or exclusion criteria due to facts occurring after screening ($n = 1$), noncompliance ($n = 1$) or due to a seronegativity for GPCR-fAABs ($n = 1$).

30 participants were randomly assigned to the RCT with one rovonaptabin arm ($n = 14$) and one placebo arm ($n = 16$). Additionally, a cross-over treatment was conducted afterwards ([Suppl. Figure S2](#)). One participant, assigned to treatment, was withdrawn due to acute SARS-CoV-2 infection (i.e., exclusion criteria, drop out) after d28. 29 participants completed the trial. Of all participants' visits, five visits of three participants were not realised due to the patients' malaise ($n = 3$) or illness ($n = 2$). Therefore, 30 participants completed V7, including the complete data set for primary outcome. Demographic and clinical data can be seen in [Table 1](#), comorbidities are listed in [Suppl. Tables S1 and S2](#) and co-medications in [Suppl. Table S3](#). Information regarding COVID-19 vaccinations can be found in [Suppl. Table S5](#). No clinically meaningful differences were observed between participants of both groups.

RCT

Analysis of the primary endpoint of the RCT yielded that the number of participants with at least one TEAE from baseline to d28 was six of 14 in the rovonaptabin group (43%) and three of 16 in the placebo group (19%), respectively. An expected temporal anticoagulatory effect directly after treatment was observed in one of 14 persons of the rovonaptabin group, showing increased INR and aPTT. The five further persons, receiving rovonaptabin, showed TEAEs with a mild or moderate severity, not related to treatment, as assessed by the clinical investigators ([Table 2](#)). The number of TEAEs was 9 in participants receiving rovonaptabin and 5 in participants receiving placebo until d28, yet this difference was not significant ($p = 0.1299$, Poisson-test). The confidence interval (CI)

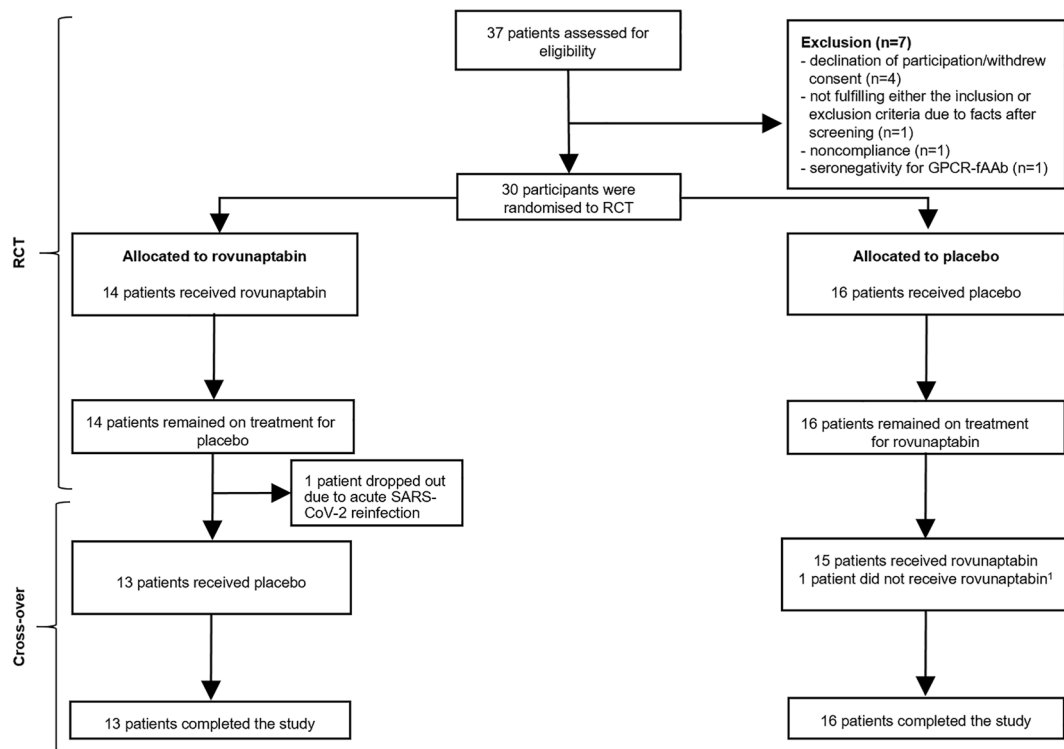


Fig. 1: Trial profile. The population included all treated participants who received at least one administration of the study drug. The population enrolled was randomised by our study pharmacy of the Universitätsklinikum Erlangen. Randomised participants passed all study inclusion or exclusion criteria. Before the infusion of either rovnaptabin (dosage: 1350 mg) or placebo PCS patients had to undergo a gargle water PCR-COVID-test in order to exclude acute SARS-CoV-2 infections. RCT: patients received either rovnaptabin/placebo on their 2nd visit with seven visits as follow-up; cross-over: patients received rovnaptabin-placebo or placebo-rovnaptabin at the 8th visit with 7 visits as follow-up. In total 14 visits were planned for each study-participant. RCT: missing visits $n = 1$ (rovnaptabin group); missing visits $n = 0$ (placebo group); cross-over: missing visits $n = 2$ (rovnaptabin-placebo); missing visits $n = 2$ (placebo-rovnaptabin); total missing visits were $n = 5$; for the primary variable TAE all documented TAEs were analysed regardless of drop-out or no missing visits. ¹One patient did not receive rovnaptabin due to an adverse event after placebo administration.

of the difference between both sequences was -14.80% ; 63.02% . The ratio of the two event rates (AE/time) was 2.29 with a 95% CI of $[0.71; 8.52]$. During treatment no clinically significant changes in vital signs were observed. One report of a serious adverse event, not related to treatment (i.e., wasp sting with the demand of hospitalisation), was recorded.

In line with its described anticoagulatory effects, rovnaptabin significantly increased the INR 2 h after infusion (Wilcoxon signed-rank test, $p = 0.0078$, [Suppl. Table S7 and Figure S1](#)). No effect on the aPTT or the prothrombin time was seen ([Suppl. Tables S6 and S8](#)), however the prothrombin time was not evaluated 2 h after infusion. The day after infusion we could no longer detect a significant increase of the INR compared to the pre-infusion timepoint, showing the anticoagulatory effects of rovnaptabin are short-lived. A summary of all reported adverse events can be seen in [Table 3](#) and [Suppl. Table S9](#).

RCT-treatment effects of rovnaptabin on fatigue, severity of fatigue and subjective rating of ME/CFS are presented in [Table 3](#). Rovernaptabin showed a significant improvement of self-reported fatigue (*FACIT Fatigue Scale*) and of the self-reported severity of fatigue (*Bell score, Fatigue Severity Scale FSS*, [Table 3](#)). Scores for each visit can be seen in [Fig. 2](#). In addition, rovnaptabin significantly improved quality of life, in particular vitality (*SF36 vitality*), general health (*SF36 general health*), and social role functioning (*SF36 social role function*, [Table 3](#)). A significant improvement of mental component summary (*SF36 MCS*) was seen under rovnaptabin treatment ([Table 3](#)). No RCT-treatment effects of rovnaptabin were observed on DSQ-PEM, walking distance (6MWT) and the subjective rating of level of exertion and dyspnoea during 6MWT (Borg scales). The two-sample Wilcoxon rank sum test for change from baseline is provided in [Suppl. Table S31](#).

Individual patterns of GPCR-fAABs can be seen in [Suppl. Tables S29 and S30](#). Seropositivity of GPCR-fAABs was 100% for positive chronotropic fAABs (100% β 2-fAAB, 7% AT1-fAAB, 0% α 1-fAAB, 0% NocifAAB) and 100% for negative chronotropic fAABs (0% MAS-AAB, 100% M2-AAB, 0% ETA-AAB) at screening (V1). In the 30 patients, both a quick ($n = 13$, 43.30%) and delayed ($n = 7$, 23.30%) effect of rovonaptabin on GPCR-fAABs were observed. A third participants' group ($n = 9$, 30%) showed a seronegativity of GPCR-fAABs at d7 (placebo first group), differing from the other participants of this group ([Suppl. Figure S4](#)). One participant (3.30%) had a persistency of GPCR-fAABs within the study period (receiving only the placebo infusion). A relapse of GPCR-fAABs (d70) was observed in one participant (3.30%).

Cross-over

Analysis of the within-participant differences yielded that eight participants of 14 (57%) of the rovonaptabin–placebo group, and nine of 16 of the placebo–rovonaptabin group (56%) reported at least one TEAE from baseline to d70, which was not significantly different ($p = 0.1352$, CI -35.54% ; 37.32% , Poisson test, co-primary endpoint). The ratio of the two event rates was 1.87 with a 95% CI of [0.84; 4.38]. Fishers exact test yielded a p -value of 0.2360. Summarising all TAE rates over time, no statistically significant differences between both groups were observed ($p = 0.1060$; Poisson test). The ratio of the two event rates was 1.84 with a 95% CI of 0.85; 4.16. Analysis of the occurrence of TAE per patient yielded no statistically significant differences between both groups ($p = 0.423$; chi-square = 0.643 with $df = 1$, McNemar test).

A significant improvement of fatigue (*FACIT Fatigue Scale*), the severity of fatigue (*Bell score*), and of ME/CFS diagnosis (CCC) was observed under rovonaptabin in the cross-over analysis. Means (SD, CI) of each score can be seen in [Suppl. Tables S10–S13](#). Additionally, a significant improvement of vitality (*SF36 vitality*) and general health (*SF36 general health*, [Table 2](#)) were observed under treatment with rovonaptabin. Scores for each visit can be seen in [Fig. 2](#), [Suppl. Figure S3](#) and [Tables S10–S28](#).

Rovonaptabin significantly increased the aPTT (Wilcoxon signed-rank test, $p = 0.0026$, [Suppl. Table S6](#) and [Figure S1](#)) and the INR 2 h after infusion (Wilcoxon signed-rank test, $p = 0.0010$). However, a significant increase in INR 2 h after placebo infusion was observed as well ($p = 0.0396$), yet no effect on the prothrombin time ([Suppl. Table S8](#)); however it was not evaluated 2 h after infusion. The day after infusion we could no longer detect significant increases in aPTT or INR compared to the pre-infusion timepoint.

	Rovonaptabin (14)	Placebo (16)	Total 30
Age [years]	41.0 $\pm$ 10.44	39.0 $\pm$ 7.92	39.9 $\pm$ 9.08
Duration of PCS [days]	1136 $\pm$ 283	1061 $\pm$ 232	1096 $\pm$ 256
Male	10 (71.4%)	10 (62.5%)	20 (66.7%)
Female	4 (28.6%)	6 (37.5%)	10 (33.3%)
Smoker	0 (0.0%)	0 (0.0%)	0 (0.0%)
Not smoking	8 (57.1%)	10 (62.5%)	18 (60.0%)
Previous smoker	6 (42.9%)	6 (37.5%)	12 (40.0%)
PCS duration [days]	1136 $\pm$ 283.0	1061 $\pm$ 232.4	1096 $\pm$ 255.5
FACIT score	18.07 $\pm$ 8.38	20.31 $\pm$ 7.26	19.28 $\pm$ 7.75
Bell score	31.43 $\pm$ 9.49	38.12 $\pm$ 16.01	35.00 $\pm$ 13.58
FSS score	6.24 $\pm$ 1.46	6.33 $\pm$ 0.44	6.29 $\pm$ 1.03
CCC score = yes	86% (12)	56% (9)	70% (21)
Vitality	17.50 $\pm$ 11.39	24.06 $\pm$ 17.05	21.00 $\pm$ 14.82
General health	25.00 $\pm$ 15.44	34.31 $\pm$ 15.07	29.97 $\pm$ 15.71
Social role functioning	33.93 $\pm$ 22.70	28.12 $\pm$ 21.65	30.83 $\pm$ 21.96
Mental component summary	32.54 $\pm$ 9.96	37.17 $\pm$ 9.90	35.01 $\pm$ 10.03
Physical function scores	46.79 $\pm$ 21.09	48.75 $\pm$ 22.40	47.83 $\pm$ 21.44
Physical role function	7.14 $\pm$ 26.73	4.69 $\pm$ 13.60	5.83 $\pm$ 20.43
Emotional role functioning	40.48 $\pm$ 43.71	67.29 $\pm$ 42.93	54.78 $\pm$ 44.67
Emotional well-being	64.00 $\pm$ 11.42	68.00 $\pm$ 12.90	66.13 $\pm$ 12.19
Physical pain	33.79 $\pm$ 23.05	47.12 $\pm$ 23.81	40.90 $\pm$ 24.02
Physical component summary	29.78 $\pm$ 6.97	30.96 $\pm$ 6.44	30.41 $\pm$ 6.61
Chalder fatigue scale	9.93 $\pm$ 2.40	9.75 $\pm$ 1.39	9.83 $\pm$ 1.90
Walking distance 6MWT	307.64 $\pm$ 114.56	335.38 $\pm$ 150.92	322.43 $\pm$ 133.65
Exertion 6MWT	13.85 $\pm$ 3.13	13.25 $\pm$ 3.70	13.52 $\pm$ 3.41
Dyspnoea 6MWT	1.15 $\pm$ 1.86	1.44 $\pm$ 1.90	1.31 $\pm$ 1.85
DSQ-PEM	29.64 $\pm$ 8.54	29.38 $\pm$ 11.23	29.50 $\pm$ 9.90

PCS: physical component summary. All values except gender, smoking status, and CCC score are given as means with SD.

Table 1: Demographic and clinical data at baseline (V2).

Discussion

The reCOVER study was an investigator-initiated, placebo-controlled randomised clinical phase IIa trial with additional cross-over to investigate safety, tolerability and clinical effects of rovonaptabin on fatigue, severity of fatigue and quality of life in PCS patients, showing a seropositivity for GPCR-fAABs. Rovonaptabin has been shown to neutralise GPCR-fAABs, both *in-vitro* and *in-vivo*.^{19,20} Data published by us and others suggest an involvement of GPCR-fAABs in the pathogenesis of PCS, leading to the hypothesis that neutralisation of such GPCR-fAABs would improve PCS symptoms.²¹ An individual treatment of a PCS patient performed by our group, showed neutralisation of GPCR-fAABs with a consecutive improvement of PCS symptoms after rovonaptabin treatment.¹⁵ The results of reCOVER confirmed these clinical observations: a significant improvement of fatigue and quality of life observed after infusion of rovonaptabin. As an intention-to-treat analysis (ITT) was performed including all randomised patients, an underestimation of the treatment effect of rovonaptabin is assumed due to inclusion of

Patient	Treatment	AE	Day of AE	Severity	Reasonable causal relation	Serious AE	System/organ class	Preferred term
RCT								
ER-02	R	Increased GOT (78 U/l) ^a	1	Mild	No	No	Investigations	Aspartate aminotransferase increased
ER-02	R	Increased GPT (140 U/l) ^a	1	Mild	No	No	Investigations	Alanine aminotransferase increased
ER-04	P	Anal vein thrombosis	20	Mild	No	No	Gastrointestinal disorders	Haemorrhoids thrombosed
ER-04	P	Influenza A	28	Moderate	No	No	Infections and infestations	Influenza
ER-06	R	Mild infection	20	Mild	No	No	Infections and infestations	Nasopharyngitis
ER-06	R	Deterioration of colitis ulcerosa	24	Mild	No	No	Gastrointestinal disorders	Colitis ulcerative
ER-10	R	Cold	16	Moderate	No	No	Infections and infestations	Nasopharyngitis
ER-11	R	Increased INR (2.89) ^b	0	Mild	Yes	No	Investigations	International normalised ratio increased
ER-11	R	Increased aPTT (61.2 s) ^c	0	Mild	Yes	No	Investigations	Activated partial thromboplastin time prolonged
ER-34	R	Depressive mood	0	Mild	No	No	Psychiatric disorders	Depressed mood
ER-34	R	Flu-like infect	24	Mild	No	No	General disorders and administration site conditions	Influenza like illness
ER-36	P	Numbness of lips, face and hand	0	Moderate	Yes	No	Nervous system disorders	Hypoesthesia
ER-36	P	Tightness of throat	0	Moderate	Yes	No	Respiratory, thoracic, and mediastinal disorders	Throat tightness
ER-37	P	Higher lipase values in bloodwork	14	Mild	No	No	Investigations	Lipase increased
Cross-over								
ER-02	RP	Seasonal infection/cold	64	Mild	No	No	Infections and infestations	Nasopharyngitis
ER-02	RP	Periodontitis	64	Mild	No	No	Infections and infestations	Periodontitis
ER-07	PR	Increased INR (>7.0) and aPTT (>180 s)	42	Moderate	Yes	No	Investigations	Coagulation test abnormal
ER-09	PR	Epidymorchitis	42	Mild	No	No	Infections and infestations	Orchitis
ER-10	RP	Cold	77	Mild	No	No	Infections and infestations	Nasopharyngitis
ER-11	RP	Gastrointestinal infection	42	Mild	No	No	Infections and infestations	Gastrointestinal infection
ER-12	PR	Palpable lymph node ^d	88	Mild	No	No	Investigations	Lymph node palpable
ER-13	PR	Increased INR (2.51)	43	Mild	Yes	No	Investigations	International normalised ratio abnormal
ER-15	PR	Increased INR (4.03)	42	Mild	Yes	No	Investigations	International normalised ratio abnormal
ER-17	RP	Angina pectoris ^e	44	Mild	No	No	Cardiac disorders	Angina pectoris
ER-17	RP	Gastric pain	48	Mild	No	No	Gastrointestinal disorders	Abdominal pain upper
ER-18	PR	Thrombophlebitis	43	Mild	No	No	Vascular disorders	Thrombophlebitis
ER-21	PR	Dyspnoea ^f	41	Moderate	No	No	Respiratory, thoracic, and mediastinal disorders	Dyspnoea
ER-28	RP	Flu-like infect	25	Moderate	No	No	General disorders and administration site conditions	Influenza like illness
ER-28	RP	Corona re-infection	38	Moderate	No	No	Infections and infestations	COVID-19
ER-33	RP	Wasp sting ^g	33	Mild ^a	No	Yes ^a	Injury, poisoning, and procedural complications	Arthropod sting
ER-33	RP	Common cold	42	Mild	No	No	Infections and infestations	Nasopharyngitis

AE: Adverse event, INR: International normalised ratio, aPTT: activated partial thromboplastin time. ER28 only received one infusion (rovunaptabin) at V2 (d0). ER36 only received one infusion (placebo) at V2 (d0). ^aRepetitive increase of GOT/GPT were known already before the trial, diagnostic reference range: GOT < 50 U/l; GPT < 50 U/l. ^bDiagnostic reference range INR: 0.85-1.15. ^cDiagnostic reference range aPTT: 28.9-38.1 s. ^dWithout previous infection. ^eWithout Troponin elevation. ^fWith oxygen saturation of 91-95%. ^gPatient had mild symptoms but stayed overnight in the hospital for monitoring due to a known wasp venom allergy, thus the AE was categorised as serious.

Table 2: Adverse events observed in study participants.

one participant, who never received rovunaptabin (so the treatment effects are conservative estimates). Rovunaptabin was observed to neutralise GPCR-fAAs in all patients. This effect might be mediated by different mechanisms, as a quick and delayed effect was observed. Of note, as this neutralisation of GPCR-

fAAs was accompanied by an improvement of self-reported fatigue and quality of life, the assumption that GPCR-fAAs exert a pathogenic effect and thus, are linked to clinical symptoms, is supported by the data of reCOVer. This observation confirmed previous data, showing that clinical features were improved after

unspecific elimination of GPCR-fAABs by immune adsorption (IA) in patients of other disorders.^{22,23} Interim results of an observational study of repeated IA in PCS patients with ME/CFS showed a decrease of β 2-AABs and an improvement of clinical PCS symptoms for seven out of 10 patients²⁴; however, no control group was included and GPCR-AAB testing was based on enzyme-linked immunosorbent assay technology. It can be assumed that a subset of GPCR-AAB mediate an additional functional activity (GPCR-fAABs). The continuous binding of GPCR-fAABs to their receptors stimulates the receptors due to their agonistic effect, decreases the number of receptors on the cell surface and their mRNA levels,²⁵ consequently affecting e.g., endothelial cells, circulating mononuclear cell function, muscles or nerves.²⁶

Although only a single infusion of rovnaptabin was applied, the patients experienced a prolonged improvement of their symptoms. As rovnaptabin has only a short serum half-life of about 9.9 min,¹² we hypothesise that the drug not only prevents binding of the GPCR-fAABs to their receptors, but in addition, it likely exerts an immunomodulating effect on B-cells and/or plasma cells. Further research is needed to delineate this assumed interaction of rovnaptabin.

Of interest a further participants' subgroup showed a disappearance of GPCR-fAABs without rovnaptabin treatment, but without an improvement of fatigue in the following weeks, compared to an improvement of fatigue in those participants, whose GPCR-fAABs were neutralised after rovnaptabin. We assume that low serum levels of highly active GPCR-fAAB still might have a functional pathogenic effect, even when their concentrations drop below the detection limit of the bioassay. The method of screening for GPCR-fAABs is based on a CM-bioassay. This laboratory method counts the difference of the beating rate of CMs after add-on of purified patients' sera compared to basal beating rate. This difference of the beating rate is transferred to a binary outcome: seropositivity of GPCR-fAAB—yes/no. As the CM-bioassay is a very labour and time intensive test, the test was done without determination of the titre.

Current treatment trends in PCS emphasise a multidisciplinary approach, involving symptomatic, supportive, and rehabilitative care, with a significant role for patient self-monitoring and active participation in their recovery. Due to the severe difference in symptoms and severity, it is likely that the cause of PCS is multifactorial, summarising several pathogenetic molecular pathways. The present study focus on a specific group of patients with PCS with fatigue as major symptom and a seropositivity of GPCR-fAAB. The data of the present study showed that rovnaptabin neutralised GPCR-fAAB, followed by a clinical improvement of fatigue, the present findings may help to develop a potential novel causal treatment for this PCS group.

Different clinical trials with PCS patients are running, differing in study protocols, patients' characteristics and outcome variables.²⁷ To the best of our knowledge this is the first study of the application of rovnaptabin in PCS patients seropositive for GPCR-fAABs. The improvement of fatigue was apparent by the FACIT fatigue scale, Bell scale, FSS, and CCC, yet not by the Chalder fatigue scale, DSQ-PEM, and Borg scales. This data demonstrates clear differences between the different fatigue questionnaires. In addition, patients did not show a significant increase of their walking distance, as measured by the 6MWT.

At its beginning, rovnaptabin was designed as thrombin binding aptamer¹⁰ with a mild-to-moderate increase in aPTT.¹² This transient change in blood anticoagulation was also observed after rovnaptabin treatment in our study. During RCT we observed 9 mild and moderate TEAEs in participants, who received rovnaptabin, and 5 mild and moderate TEAEs in patients, who received placebo. Overall number of TEAE rate was not significantly increased in participants of the treatment group compared to placebo, possibly also due to the low number of participants. The rate and severity of TEAE, observed in reCOVER is in line with previous studies.^{12,28,29}

While reCOVER showed a positive effect of rovnaptabin, a phase II clinical trial by Berlin Cures GmbH (BLOC) seemed not to show a superior effect compared to placebo in its primary endpoint, as announced in a press release, yet data of participant characteristics and outcomes have not been published yet.³⁰

RCT	Improvement of the score	95% CI	SD	p-value	t-value
FACIT	2.91	0.17–5.65	1.38	0.0378	2.10
Bell	7.54	3.43–11.64	2.07	0.0004	3.64
FSS	–0.43	–0.75 to –0.11	0.16	0.0088	–2.66
SF36					
Vitality	6.65	3.38–9.93	1.65	0.0001	2.97
General health	4.86	1.04–8.67	1.92	0.0130	2.52
Social functioning	6.46	0.31–12.61	3.10	0.0396	2.08
Mental component	2.73	0.19–5.28	1.28	0.0357	2.13
Crossover					
FACIT	2.73	0.92–4.54	0.92	0.0033	2.97
Bell	3.43	0.84–6.02	1.31	0.0097	2.61
CCC	–0.80	–1.58 to –0.01	0.40	0.0467	–1.99
SF36					
Vitality	3.29	0.53–6.05	1.40	0.0196	2.35
General health	3.41	0.81–6.00	1.32	0.0104	2.58

CI: Confidence interval, SD: Standard deviation. RCT analysis was done between the rovnaptabin and placebo arm; the treatment effect was calculated as the difference between both treatment groups, adjusted for baseline. The scores showing a significant improvement after treatment with rovnaptabin are shown: Fatigue scores—FACIT, Bell, FSS; quality of life—SF36: Vitality, general health, social functioning, mental component summary. The unit of the treatment effect is the unit of the corresponding questionnaire. Cross-over differences show the in-patients differences for each score, for which a significant improvement was observed after treatment with rovnaptabin compared to baseline.

Table 3: Treatment effect of rovnaptabin.

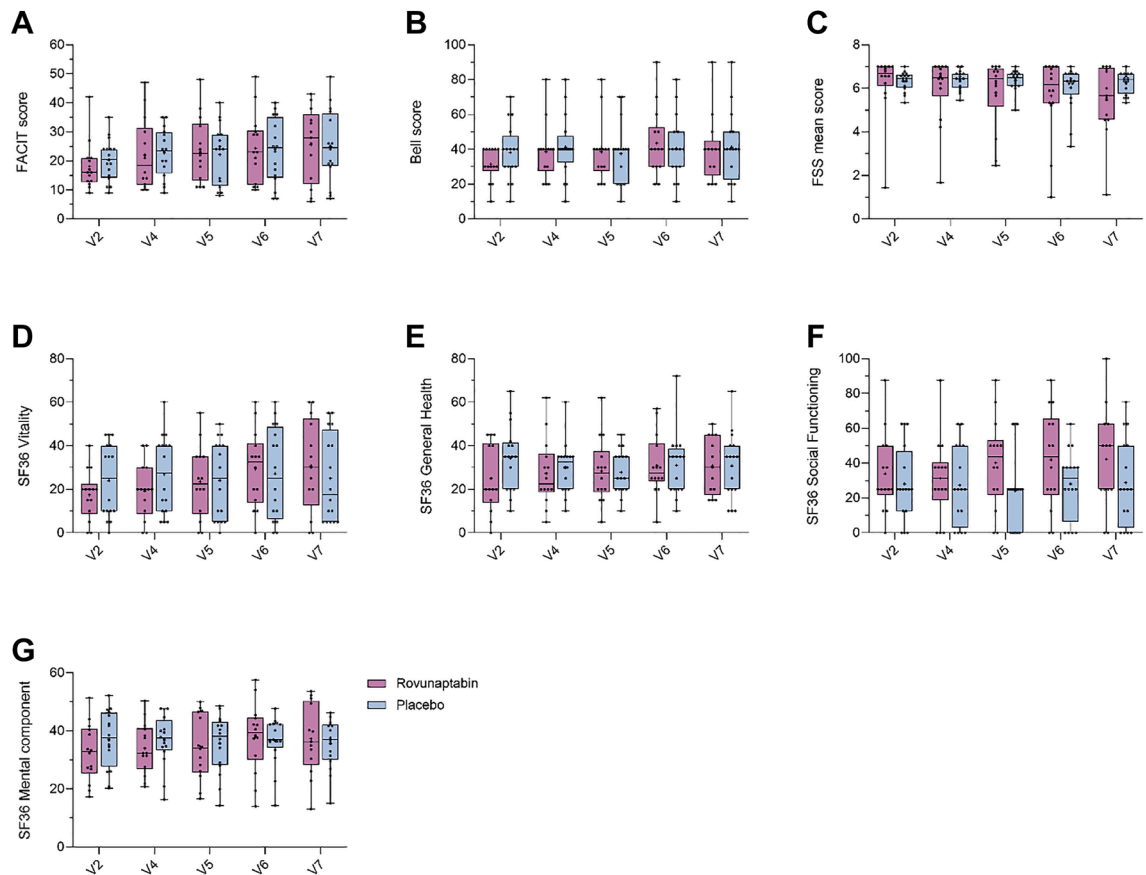


Fig. 2: Fatigue scores assessed at V2, V4, V5, V6, and V7. (A) FACIT scores, (B) Bell scores, and (C) FSS mean scores of study participants. Depicted are medians with IQR and ranges. Means are shown as crosses. (D) SF36 vitality scores, (E) SF36 General Health scores, (F) SF36 social functioning scores, and (G) SF36 mental component summary scores of study participants. Depicted are medians with IQR and ranges. Means are shown as crosses. Sequence A n V2–V6 = 14, n V7 = 13. Sequence B n V2–7 = 16. One patient in sequence A only received the first infusion (rovunaptabin) at V2 (ER28). One patient in sequence B only received the first infusion (placebo) at V2 (ER36).

ReCOVer is the first RCT of rovunaptabin in PCS patients with a seropositivity of GPCR-fAAb, showing encouraging results. The additional cross-over design enabled analysis of inter-patients' differences in validated self-reported patients' outcomes. As no epidemiology data on the presence of GPCR-fAAb are available up to date, the number of participants was low, the study cohort represent an European cohort and recruitment was random as patients registered actively for a potential study participation, the data of the present study cannot be generalised. Further limitations were the use of self-reported questionnaires only, and the lack of established biomarkers except for GPCR-fAAb. Patients had to complete a larger number of questionnaires, as the effect of rovunaptabin on the outcome variables was not known. As the follow-up duration was limited to a maximum of 91 days, the present results of reCOVer encourage a subsequent clinical phase II b, investigating the doses and the effects in a larger patients'

group over a longer follow-up period, as well as subsequent phase III trial.

The prevalence of PCS seems to be linked to several risk factors and even the number of re-infections,³¹ thus, it can be assumed that the total number of patients with PCS will still increase in future, affecting individual life, public health care system and economy.^{32,33} The clinical trial reCOVer is the first RCT of rovunaptabin in PCS patients seropositive for GPCR-fAAb, showing a neutralisation of GPCR-fAAb, followed by a significant improvement of fatigue, severity of fatigue and quality of life compared to placebo. Next to an expected temporal anticoagulatory effect directly after treatment, participants of the rovunaptabin group showed mild to moderate TEAEs, compared to the placebo group, yet neither with a statistical significance, nor with treatment related SAEs. The data of the present clinical trial suggest that rovunaptabin might offer a novel approach for a curative and causal therapy for an autoimmune subgroup of patients with PCS.

Contributors

Conceptualisation: BH, CMA, TH, SM, MH, MG, JR.
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 Writing – review & editing: All authors.

Data sharing statement

Anonymised data are available from the corresponding author upon request following publication by reasonable formal request from qualified researchers, subject to a signed data sharing agreement and in compliance with the requirements of the funding bodies and institutions.

Declaration of interests

BH and MH declare conflict of interests; BH: German Federal Ministry of Education and Research (BMBF) 01EP2108A, reCOVer, research grant; German Federal Ministry of Education and Research (BMBF) 01EO2105, iIMMUNE_ACS, research grant; German Research Foundation (DFG) 401821119/GRK2504, research grant; German Federal Ministry of Education and Research (BMBF) 01EJ2409, FAME, research grant; The Bavarian Health and Food Safety Authority (LGL) K1-2497-Projekte-22-V9-D51632/2022, uncover, research grant; German Federal Ministry of Health (BMG) Post-COVID Meeting, Berlin 25-11-2024; Member of the German S1 Guideline Long-/Post-COVID; MH: German Federal Ministry of Education and Research, BMBF 01EJ2409, FAME, Subproject no. 5, research grant; the other authors declare no competing interests.

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No patient or public involvement was involved in the different trial stages.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.eclim.2025.103358>.

References

- Soriano JB, Murthy S, Marshall JC, Relan P, Diaz JV, WHO Clinical Case Definition Working Group on Post-COVID-19 Condition. A clinical case definition of post-COVID-19 condition by a Delphi consensus. *Lancet Infect Dis*. 2022;22(4):E102–E107.
- Parotto M, Gyongyosi M, Howe K, et al. Post-acute sequelae of COVID-19: understanding and addressing the burden of multisystem manifestations. *Lancet Respir Med*. 2023;11(8):739–754.
- Greenhalgh T, Sivan M, Perkowski A, Nikolich JZ. Long COVID: a clinical update. *Lancet*. 2024;404(10453):707–724.
- Proal AD, VanElzakker MB. Long COVID or post-acute sequelae of COVID-19 (PASC): an overview of biological factors that may contribute to persistent symptoms. *Front Microbiol*. 2021;12:698169.
- Kell DB, Laubscher GJ, Pretorius E. A central role for amyloid fibrin microclots in long COVID/PASC: origins and therapeutic implications. *Biochem J*. 2022;479(4):537–559.
- Ackermann M, Anders HJ, Bilyy R, et al. Patients with COVID-19: in the dark-NETs of neutrophils. *Cell Death Differ*. 2021;28(11):3125–3139.
- Castanares-Zapatero D, Chalon P, Kohn L, et al. Pathophysiology and mechanism of long COVID: a comprehensive review. *Ann Med*. 2022;54(1):1473–1487.
- Wallukat G, Hohberger B, Wenzel K, et al. Functional autoantibodies against G-protein coupled receptors in patients with persistent Long-COVID-19 symptoms. *J Transl Autoimmun*. 2021;4:100100.
- Szewczykowski C, Mardin C, Lucio M, et al. Long COVID: association of functional autoantibodies against G-protein-coupled receptors with an impaired retinal microcirculation. *Int J Mol Sci*. 2022;23(13):7209.
- Bock LC, Griffin LC, Latham JA, Vermaas EH, Toole JJ. Selection of single-stranded DNA molecules that bind and inhibit human thrombin. *Nature*. 1992;355(6360):564–566.
- Haberland A, Wallukat G, Schimke I. Aptamer binding and neutralization of beta1-adrenoceptor autoantibodies: basics and a vision of its future in cardiomyopathy treatment. *Trends Cardiovasc Med*. 2011;21(6):177–182.
- Becker NP, Haberland A, Wenzel K, et al. A three-part, randomised study to investigate the safety, tolerability, pharmacokinetics and mode of action of BC 007, neutraliser of pathogenic autoantibodies against G-protein coupled receptors in healthy, young and elderly subjects. *Clin Drug Investig*. 2020;40(5):433–447.
- Cocks K, Torgerson DJ. Sample size calculations for pilot randomized trials: a confidence interval approach. *J Clin Epidemiol*. 2013;66(2):197–201.
- Berger VW, Ivanova A, Knoll MD. Minimizing predictability while retaining balance through the use of less restrictive randomization procedures. *Stat Med*. 2003;22(19):3017–3028.
- Hohberger B, Harrer T, Mardin C, et al. Case report: neutralization of autoantibodies targeting G-protein-coupled receptors improves capillary impairment and fatigue symptoms after COVID-19 infection. *Front Med (Lausanne)*. 2021;8:754667.
- Team RC. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing; 2024. <https://www.R-project.org/>.
- Newcombe RG. Interval estimation for the difference between independent proportions: comparison of eleven methods. *Stat Med*. 1998;17(8):873–890.
- Senn S. *Cross-over Trials in Clinical Research*. John Wiley & Sons, Ltd; 2002.
- Haberland A, Holtzhauer M, Schlichtiger A, et al. Aptamer BC 007 - a broad spectrum neutralizer of pathogenic autoantibodies against G-protein-coupled receptors. *Eur J Pharmacol*. 2016;789:37–45.
- Haberland A, Wallukat G, Dahmen C, Kage A, Schimke I. Aptamer neutralization of beta1-adrenoceptor autoantibodies isolated from patients with cardiomyopathies. *Circ Res*. 2011;109(9):986–992.
- Cabral-Marques O, Moll G, Catar R, et al. Autoantibodies targeting G protein-coupled receptors: an evolving history in autoimmunity. Report of the 4th international symposium. *Autoimmun Rev*. 2023;22(5):103310.
- Junemann A, Hohberger B, Rech J, et al. Agonistic autoantibodies to the beta2-adrenergic receptor involved in the pathogenesis of open-angle glaucoma. *Front Immunol*. 2018;9:145.
- Muller J, Wallukat G, Dandel M, et al. Immunoglobulin adsorption in patients with idiopathic dilated cardiomyopathy. *Circulation*. 2000;101(4):385–391.
- Stein E, Heindrich C, Wittke K, et al. Observational study of repeat immunoadsorption (RIA) in post-COVID ME/CFS patients with elevated ss2-adrenergic receptor autoantibodies-an interim report. *J Clin Med*. 2023;12(19):6428.
- Podlowski S, Luther HP, Morwinski R, Muller J, Wallukat G. Agonistic anti-beta1-adrenergic receptor autoantibodies from cardiomyopathy patients reduce the beta1-adrenergic receptor expression in neonatal rat cardiomyocytes. *Circulation*. 1998;98(22):2470–2476.
- Sotzny F, Filgueiras IS, Kedor C, et al. Dysregulated autoantibodies targeting vaso- and immunoregulatory receptors in Post COVID

- Syndrome correlate with symptom severity. *Front Immunol.* 2022;13:981532.
- 27 Yong SJ, Halim A, Halim M, et al. Experimental drugs in randomized controlled trials for long-COVID: what's in the pipeline? A systematic and critical review. *Expert Opin Investig Drugs.* 2023;32(7):655–667.
- 28 Griffin LC, Tidmarsh GF, Bock LC, Toole JJ, Leung LL. In vivo anticoagulant properties of a novel nucleotide-based thrombin inhibitor and demonstration of regional anticoagulation in extra-corporeal circuits. *Blood.* 1993;81(12):3271–3276.
- 29 DeAnda A Jr, Coutre SE, Moon MR, et al. Pilot study of the efficacy of a thrombin inhibitor for use during cardiopulmonary bypass. *Ann Thorac Surg.* 1994;58(2):344–350.
- 30 Website of Berlin Cures. <https://www.berlincures.com/en/news/phase-2-long-covid-results>; 2024.
- 31 Bowe B, Xie Y, Al-Aly Z. Acute and postacute sequelae associated with SARS-CoV-2 reinfection. *Nat Med.* 2022;28(11):2398–2405.
- 32 Carlile O, Briggs A, Henderson AD, et al. Impact of long COVID on health-related quality-of-life: an OpenSAFELY population cohort study using patient-reported outcome measures (OpenPROMPT). *Lancet Reg Health Eur.* 2024;40:100908.
- 33 Taher MK, Salzman T, Banal A, et al. Global prevalence of post-COVID-19 condition: a systematic review and meta-analysis of prospective evidence. *Health Promot Chronic Dis Prev Can.* 2025;45(3):112–138.